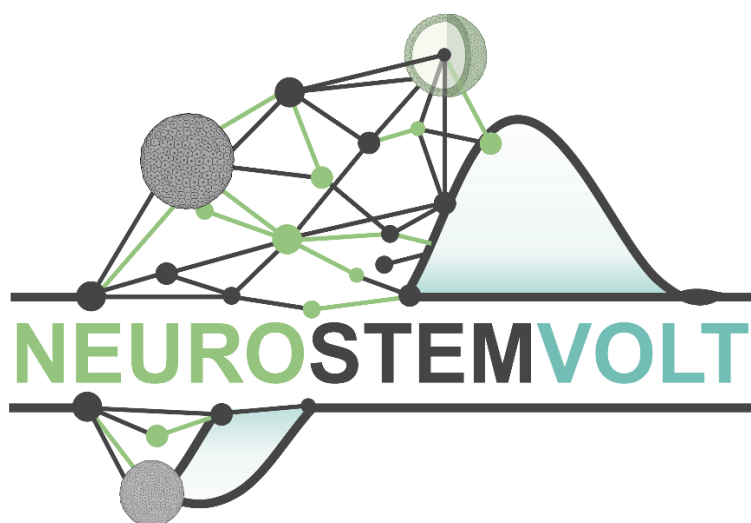




Neurotransmitter dynamics
from **Stem** cell-derived models
by **Volt**ammetry

User Manual

Version 1.1.1
macOS • Windows

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1. Introduction

1.1 Overview of NeuroStemVolt

NeuroStemVolt is a desktop application for the analysis of fast-scan cyclic voltammetry (FSCV) data collected from neuronal cultures derived from induced pluripotent stem cells (iPSCs). It provides an integrated environment for loading raw voltammetric recordings, applying signal pre-processing, detecting and characterizing neurotransmitter release events, and exporting results for downstream statistical analysis.

The software is designed for neuroscience researchers who need a reproducible, visual, and configurable workflow for turning raw electrochemical time-series datasets into quantitative measures of neurotransmitter release and reuptake, without requiring custom scripting.

1.2 Intended Use & Target Users

NeuroStemVolt is intended for use by researchers, technicians, and students working with FSCV data in preclinical or *in vitro* neuroscience settings, including studies of serotonin, histamine and other electroactive neurotransmitters. It is intended for research use only.

- Electrochemistry and neurobiology labs analyzing evoked neurotransmitter release
- Pharmacology groups profiling drug effects on release and clearance
- Core facilities processing large batches of FSCV recordings

1.3 Key Capabilities Summary

- Loading, visualization, and interactive exploration of FSCV time-series measurements
- Configurable pre-processing pipeline (baseline correction, filtering, smoothing)
- Adaptive single- and multi-peak detection with a guaranteed fallback result
- Quantification of spontaneous and stimulus-evoked release and reuptake kinetics
- Batch processing across multiple replicates
- Export of plots, tables, settings, and experiment logs to standard file formats

1.4 System Requirements & Supported Platforms

NeuroStemVolt runs as a standalone application on macOS, Windows, and Linux. No separate Python installation or coding is required.

- macOS 11 or later (Intel or Apple Silicon)
- Windows 10 or later (64-bit)
- Linux (recent Ubuntu/Debian-based or equivalent distribution)
- Minimum 8 GB RAM recommended for large or multi-file batch datasets

2. Installation & Getting Started

2.1 Downloading the Application

Download the appropriate distribution package for your operating system from

<https://github.com/Hashemi-Lab/NeuroStemVolt>.

Three distributions are provided: a macOS application bundle, a Windows executable, and the Python source code.

2.2 Installation Instructions

2.2.1 macOS (NeuroStemVolt.app)

- Download and unzip `NeuroStemVolt-mac-v1.1.1.zip`.
- Double-click `NeuroStemVolt.app` to launch.
- If macOS says the app is from an unidentified developer:
 - Right-click `NeuroStemVolt.app`
 - Select Open → Open (*one time only*)
- If macOS says the app is “damaged”, open Terminal and run:

```
xattr -cr /path/to/NeuroStemVolt.app
```

2.2.2 Windows (NeuroStemVolt.exe)

- Download `NeuroStemVolt-windows-v1.1.0.exe`.
- Double-click to launch.
- If Windows shows a security prompt:
 - Click More info
 - Select Run anyway

2.3 Launching the Application

On first launch, NeuroStemVolt opens to its main window with no dataset loaded. The application does not require an internet connection or login and stores its configuration locally.

2.4 Interface Overview

The interface is organized into three consecutive windows, reflecting the overall workflow from data loading through to result export. Each window is described below, along with the pop-up windows used for specific configuration tasks.

2.4.1 Window 1: Experiment Settings and Data Loading

The first window is used to define experiment settings (Section 3) and to load datasets and replicates (Section 4). Menu items are arranged along the top of the window, providing access to experiment setup and data-loading functions. Below the menu, a list of loaded replicates shows each dataset that has been imported into the current session.

2.4.2 Window 2: Visualization, Pre-Processing, and Peak Detection

The second window is used to visualize color plots, current-time (I-T) profiles, and cyclic voltammograms, and to apply and review pre-processing and peak detection (Sections 5 and 6). Menu items are arranged along the left side of the window, with the corresponding visualizations and plots displayed alongside them. Results and annotated plots from this window can also be exported here.

2.4.3 Window 3: Plotting and Interpretation of Results

The third window is used for plotting, visualizing, and exporting analysis results (Section 7). As with Window 2, menu items are arranged along the left side of the window, with the corresponding result plots displayed alongside them.

2.4.4 Pop-Up Windows

Several features open in additional pop-up windows rather than within the three main windows. For example, experiment settings (Section 3) and filter/smoothing options (Section 5) are each configured in a dedicated pop-up window, which closes or applies its settings back to the main workflow once confirmed.

Note: It is possible to move back and forth between all three windows at any stage of the analysis. Changing a parameter in an earlier window (e.g., experiment settings in Window 1) will automatically update the downstream visualizations and analyses in Windows 2 and 3.

2.5 Example Dataset

An example dataset is available for download from the same GitHub repository as the software (Section 2.1), allowing new users to explore NeuroStemVolt's features without needing their own recordings.

The example dataset includes:

- Two replicates each of a **Control** condition and an **SSRI** treatment condition
- A time series of 16 files per replicate, with 3 measurements recorded before treatment addition (i.e., Files Before Treatment = 3; Section 3.5)
- The raw FSCV data, provided as tab-separated .txt files organized into replicate folders (Section 4.1, 4.2)
- The corresponding analysis output, including all plots and tables referenced throughout this manual (Sections 7, 8, Appendix B)

3. Setting Experiment Parameters

Before loading or analyzing data, NeuroStemVolt requires the experiment parameters to be defined so that raw current files can be correctly interpreted as voltage and time. These parameters describe how the data were acquired and how the experiment should be analyzed.

Note: To begin setting experiment parameters, click Experiment Settings in Window 1. This opens the corresponding pop-up window, in which the waveform, experiment mode, measurement, and stimulation parameters described below are configured.

3.1 Waveform Configuration

The **Waveform** panel provides a drop-down menu for selecting the voltage waveform applied during data acquisition, which determines how raw current data are reconstructed into voltages/cyclic voltammograms. Three options are available:

- **5-HT (Serotonin):** applies the Jackson waveform (Jackson, Frost & Moghaddam, *Anal. Chem.* 1995; DOI: [10.1021/ac00102a015](https://doi.org/10.1021/ac00102a015)), used for serotonin detection.
- **HA (Histamine):** applies the waveform described in Samaranayake et al. (DOI: [10.1039/c5an00313j](https://doi.org/10.1039/c5an00313j)), used for histamine detection.
- **Non-specific:** does not convert data points into voltages. This option allows the software to be used more broadly with waveforms outside the two predefined analytes.

Note: When Non-specific is selected, visualization of cyclic voltammograms (Window 2) is not available, since the voltage axis cannot be reconstructed.

3.2 Experiment Mode Selection

The **File Type** panel provides a drop-down menu for selecting the experiment mode, which determines how NeuroStemVolt segments and analyzes the recording. Available modes correspond to the peak-detection modes described in Section 6, and should be selected based on whether a single, well-defined release event (e.g., external stimulation; **Single Peak**) or multiple discrete events (e.g., spontaneous activity; **Multi-Peak**) are expected per recording.

3.3 Measurement Parameters

Measurement parameters describe the recording itself and are used throughout pre-processing and analysis.

- **Acquisition Frequency (Hz):** sampling rate of the acquisition system, in Hertz (Hz).
- **File Length (seconds):** total duration, in seconds, of each recorded file.
- **Time Between Files (minutes):** interval, in minutes, between each stimulation or recording session.

3.4 Oxidation Peak Definition

Peak Position defines the expected position of the signal peak on the voltage axis. This position determines which I-T profile is extracted from the color plot for subsequent analysis: the software reads out current values along the oxidation-peak voltage over time, rather than analyzing the full color plot directly.

Note: You may enter an approximate value and adjust it later after identifying the actual peak.

3.5 Treatment Parameters

This section defines the treatment condition applied during the experiment, allowing drug or treatment addition to be tracked against the recorded files for later drug response profiling (Section 7.5).

- **Treatment:** name of the treatment applied.
- **Files Before Treatment:** number of recording files acquired before applying the treatment. First treatment file will be displayed as timepoint 0 min.

3.6 Stimulation Parameters

For evoked-release experiments, stimulation parameters define when and how the stimulus was delivered so that stimulus-evoked peaks can be correctly aligned and quantified.

When **Single Peak** is selected as the experiment mode (Section 3.2), clicking **OK** at the end of the Experiment Settings pop-up opens a second, dedicated **Stimulation Parameters** pop-up window, in which the following parameters are defined:

- **Start:** start time of stimulation, in seconds.
- **Frequency:** frequency of stimulation pulses, in Hz.
- **Amplitude:** amplitude of stimulation current, in μA .
- **Pulses:** total number of stimulation pulses.

3.7 Define Output Folder

This section specifies the output folder to which exported plots, tables, settings, and experiment logs (Section 8) will be saved for the current session.

- **Output Folder:** text field displaying the currently selected output folder path.
- **Browse button:** opens a folder-selection dialog, allowing the user to navigate to and choose the desired output folder.

Note: Experiment parameters can be changed at any stage of the analysis by returning to Window 1. Adjusting these parameters after loading data or running analysis will update the corresponding time/voltage calculations accordingly.

Note: All experiment settings defined in Section 3 are stored locally and will automatically be applied the next time the software is opened, so they do not need to be re-entered for every session.

4. Loading and Viewing FSCV Data

4.1 Supported Data Formats

NeuroStemVolt accepts tab-delimited `.txt` files as input. Each file should contain the raw FSCV recording in this format for the software to correctly parse and load the dataset.

Within each file, data should be structured as follows:

- **Columns:** time points.
- **Rows:** voltage sweep (i.e., the current values recorded across the applied voltage ramp at each time point).

Note: Files exported in other formats (e.g., comma-separated `.csv`, or proprietary acquisition-system formats) should be converted to a tab-delimited `.txt` file, arranged with this row/column structure, before loading into NeuroStemVolt.

Note: We recommend labeling time-series data consistently across replicates analyzed in batch (e.g., a shared naming convention with sequential numbering) to prevent sorting issues when exporting results as `.csv`.

4.2 Importing Datasets

Use the **Load Replicates** button to select data for analysis. Each replicate is loaded as a folder of time-series data, containing one tab-delimited `.txt` file per timepoint. To add additional replicates, click **Load Replicates** again. Once loaded, each folder is listed individually in the replicate list.

Note: Batch processing of replicates from different treatment conditions is not currently possible. All replicates loaded together in a single batch should share the same treatment condition and experiment settings (Section 3.5).

Note: Individual replicate folders can be removed by selecting them in the list and pressing **Backspace**.

4.3 Visualizing Raw Color Plot

Once all replicates have been loaded, click **Next** at the bottom of Window 1 to move to Window 2 for visualization.

The color plot renders the full voltammetric recording as a two-dimensional heat map, with time on the x-axis, applied voltage on the y-axis, and measured current response represented by color scale. This view is useful for identifying redox events and confirming that detected peaks correspond to expected electrochemical signatures.

By default, the color scale is set automatically based on the maximum current value within each individual color plot. The **Visualization** panel allows an upper **manual color limit** to be set (in nA), so that the color scale can instead be fixed across different color plots or replicates.

Note: When manual color limits are set, data points outside the defined scale (i.e., exceeding the maximum or below the minimum) will be displayed in the same color as the maximum/minimum value, rather than being shown as out-of-range.

The **Navigation** panel of Window 2 provides:

- **Drop-down menus** to select the replicate and timepoint currently displayed.
- **Previous / Next buttons** to step through replicates or timepoints sequentially, without needing to use the drop-down menus each time.

4.4 Current-Time Profile Display

The current-time (I-T) profile is displayed by default below the color plot in Window 2. It is extracted from the color plot at the oxidation peak position defined in Section 3.4, giving the current response over time at the analyte's expected oxidation potential.

The displayed I-T profile updates automatically along with the color plot as the replicate or timepoint selection is changed using the **Navigation** panel (Section 4.3).

The **Visualization** panel also provides the option to display a stimulation (**Stim**) **Reference** overlaid on the I-T profile, showing the stimulus start and duration. By default, this reference is based on the stimulation parameters defined in Section 3.6, but it can be manually adjusted within the visualization panel if needed.

Note: Manually adjusting the stimulation reference in this panel affects only its display on the I-T profile; to permanently change the stimulation parameters used for analysis, update them in Section 3.6 via Window 1.

4.5 Visualizing Cyclic Voltammograms

The cyclic voltammogram (CV) view can be activated in the visualization panel via the **Show CV Plot** option. This view plots current against applied voltage, reproducing the characteristic CV shape used to confirm analyte identity.

This view makes the most sense after peak detection (Section 6) has been run: once a peak has been detected, the CV shown corresponds to the detected peak position, allowing the shape of the actual event to be inspected directly.

The visualization panel also provides an option to add a **CV reference line**, at a manually set voltage. This adds an additional line to both the CV plot and the color plot, allowing the detected peak position to be visually compared against a theoretical or expected peak voltage.

Note: The CV reference line is a visual comparison aid only — it does not change the oxidation peak position defined in Section 3.4 or affect profile extraction or analysis.

5. Data Pre-Processing

5.1 Overview of Processing Pipeline

NeuroStemVolt provides a configurable pre-processing pipeline that can be applied to the raw color plots before event detection and analysis. All pre-processing options are available in a pop-up window, opened by clicking **Filter Options** in the **Processing** panel. Processing steps can be enabled independently and are applied in the order presented in this section. The color plot and I-T profile update automatically to reflect whichever processing options are enabled.

The Filter Options pop-up window provides the following options:

- **Apply**: applies all currently enabled filters directly to the color plot, without closing the window, allowing the result to be reviewed before deciding on further changes.
- **Reverse Changes**: reverts the data back to raw (unprocessed), allowing different filter strategies to be tried from a clean starting point.
- **OK**: applies all currently enabled filters and closes the window.
- **Cancel**: closes the window without applying any filters.

Note: The **Reverse Changes** button in the **Processing** panel reverts all processing steps performed on the data, not only filtering, including peak detection (Section 6) and normalization (Section 6.7).

5.2 Background Subtraction

Background subtraction removes the non-faradaic (background) charging current from the signal, isolating the faradaic current associated with analyte oxidation and reduction.

The signal is averaged over a specified start and end segment, defined as time points (usually) at the beginning of the profile, and this mean is subtracted from the entire recording.

Note: We do not recommend enabling this option if background subtraction has already been applied in the FSCV acquisition software.

5.3 Rolling Mean

Rolling mean smooths each voltage profile by computing a moving average over a sliding window of points. The **Window Size** parameter sets how many consecutive samples are included in each average; larger windows yield smoother profiles but can blur sharp features.

5.4 Butterworth Filter

A Butterworth low-pass filter can be applied to attenuate noise while preserving the underlying waveform shape. The **Order (p)** controls the steepness of the filter roll-off, while **cx** and **cy** set the cutoff frequencies (Hz) in the time and voltage dimensions, respectively. A lower cx value results in more smoothing along the time axis.

5.5 Savitzky-Golay Filter

The Savitzky-Golay filter fits a local polynomial over each segment of the data to smooth noise. The **Window** size sets how many points are used per fit, while the polynomial **Order** controls how closely the fit can follow rapid changes.

5.6 Baseline Correction

Baseline correction addresses cases where the recording does not start at 0.0 nA prior to the onset of stimulation. It computes the mean current across the period prior to stimulation (based on the parameter set in Section 3.6) and subtracts this baseline value from the entire signal, so that the profile begins at zero before the stimulus-evoked response.

Note: This option is not available for **Multi-Peak** files (see section 3.2).

5.7 Artifact Removal

Artifact removal detects and removes sudden, large electrical noise spikes, sharp jumps in current that appear in only one or a few scans and stand out massively from the surrounding signal.

- **Threshold:** controls how many times larger a jump must be, relative to the median, to be flagged as an artifact.
- **Pad:** sets how many neighboring scans are also zeroed out around each detected artifact.
- **Max Scans:** sets the widest a single artefact is allowed to be, in scans. A spike usually produces two jumps: one as the signal leaves the baseline, and one as it returns. If these two jumps are no more than Max Scans apart, they are treated as one artefact and everything between them is interpolated; if they are further apart, they are treated as separate one-scan events. Leaving this blank uses twice the acquisition frequency, i.e. two seconds of recording.

Note: This feature is not designed for stimulation artifacts. It is intended for random electrical interference that contaminates isolated scans. If no such noise is detected, the data is returned unchanged.

5.8 Invert Data

Invert Data flips the sign of the signal. Use this option if your data was recorded or exported by the FSCV acquisition software with reversed polarity.

Note: Multiple pre-processing steps can be combined; the processed profile updates interactively so that the effect of each step can be evaluated before proceeding to event detection.

6. Event and Peak Detection

6.1 Overview of Adaptive Peak Validation

NeuroStemVolt uses an adaptive peak validation approach tailored to neurotransmitter-like signals, distinguishing genuine release events from noise based on signal shape, amplitude, and timing relative to the stimulus. Detection behavior depends on the experiment mode selected in Section 3.2.

6.2 Single-Peak Mode

When **Single Peak** mode is selected (Section 3.2), one primary release event per recording is identified, appropriate for experiments in which a single, well-defined evoked response is expected.

The Filter Options pop-up window (Section 5.1), accessed via the Processing panel, provides two parameters to control detection:

- **Prominence (%)**: the required peak height relative to baseline.
- **Min Height (nA)**: the minimum absolute peak amplitude.

As a fallback, if no peak meets these criteria, the maximum signal value within each file is identified as the peak, and can be manually adjusted afterward if needed (Section 6.4).

6.3 Multi-Peak Mode

When **Multi-Peak** mode is selected (Section 3.2), multiple discrete events within a recording can be identified, appropriate for experiments involving repeated stimulation, spontaneous release events, or multiple analytes. Each detected peak is individually annotated, analyzed (e.g., peak amplitude), and exported.

When **Multi-Peak** mode is selected, the Peak Detection panel provides the following parameters to control sensitivity and event separation:

- **Prominence (%)**: the required peak height relative to baseline.
- **Min Height (nA)**: the minimum absolute peak amplitude.
- **Max Peaks**: the maximum number of peaks to detect within a recording.
- **Min Distance (s)**: the minimum time separation required between two detected peaks.

An **Advanced** section provides additional parameters for finer control:

- **Min Peak Width (scans)**: the minimum width, in scans, a peak must span to be validated.
- **Rise Window min (s) / max (s)**: the minimum and maximum duration allowed for the rise phase of a peak.
- **Decay Window min (s) / max (s)**: the minimum and maximum duration allowed for the decay phase of a peak.

A **Reset Advanced** to Defaults button restores all Advanced parameters to their default values.

6.4 Manual Review and Adjustment of Detected Events

The Peak Editing panel provides options to manually review and adjust detected peaks.

Note: We recommend performing manual adjustments before normalization (Section 6.5). This is especially important, if the peak in the first file of a replicate needs to be adjusted, since this file serves as the normalization reference.

Activating **Edit on Plot** in the Peak Editing panel allows the user to change a peak's position (Single Peak mode) or add new peaks (Multi-Peak mode) by clicking directly on the desired position in the color plot or I-T profile.

The **Edit Peaks** option opens a separate pop-up window listing all detected or manually added peaks. Peaks are identified by their position on the time axis (**Index**). The position of the peak currently marked (**active**) can be changed by editing its Index in the list. Peaks can also be added or removed from this window (Multi-Peak mode only).

6.5 Normalization

Normalization scales detected peak amplitudes and profiles relative to the first file in each replicate's time series. It is applied by clicking **Normalize** in the Processing panel. Once applied, a checkmark is displayed to indicate that normalization is active.

7. Plotting and Interpretation of Results

Plotting and analysis options are grouped according to whether they apply across all experiment modes, or are specific to Single Peak or Multi-Peak analysis (Section 3.2). Section 7.1 covers plots available regardless of experiment mode; Sections 7.2 and 7.3 cover plots specific to Single Peak and Multi-Peak experiments, respectively.

In addition to the plots described below, the following metrics are available as `.csv` exports (Section 8.2):

- **Area under the curve (AUC)**
- **I-T profiles**
- **Reuptake curves**
- **Fitting parameters**

7.1 Analyses For All Experiment Modes

The following plots are available regardless of experiment mode:

- **Mean Amplitude Over Experiments:** plots the mean peak amplitude across all loaded replicates, allowing comparison of average response magnitude between conditions or treatments.
- **Individual Amplitudes Per Replicate:** plots the peak amplitude for each individual replicate, allowing inspection of variability across replicates within a condition.

7.2 Analyses For Single Peak Experiments

The following plots are available specifically for Single Peak experiments:

- **Tau - Time:** plots the decay constant (τ), extracted from the exponential decay fit of the detected peak, across the time course of the experiment, allowing reuptake kinetics to be tracked over time (e.g., before and after treatment).
- **Exponential Decay Fitting:** displays the fitted exponential decay curve overlaid on the detected peak's decay phase, allowing visual assessment of fit quality alongside the resulting decay constant.
- **Mean I-T Profiles:** plots the mean I-T profile for every timepoint, overlaid on a single plot and colored by treatment status, with the first file after treatment highlighted for reference, allowing visual comparison of profile shape before vs. after treatment.

7.3 Analyses for Multi-Peak Experiments

The following plots are available specifically for Multi-Peak experiments:

- **Peak Frequencies:** plots the frequency of detected peaks (e.g., number of events per unit time), allowing comparison of event rate across replicates or conditions.
- **Amplitude per Time Window:** averages peak amplitudes within user-defined time windows and plots the resulting mean amplitude over the course of the experiment. Clicking this option opens an **Averaging Window** pop-up, in which the window size (in seconds) is set; the default is 60 seconds.

8. Exporting Results

All exports are saved to the output folder defined in Section 3.7.

Note: Existing files in the output folder will be overwritten without notice.

8.1 Saving Plots

Plots can be exported as image files for inclusion in presentations, publications, or lab records.

- **From Window 2:** I-T profiles, color plots, cyclic voltammograms (CVs)
- **From Window 3:** current plot, all plots (Section 7)

All plots are saved as both `.png` and `.svg` files.

8.2 Exporting Data

Raw and processed data, along with results from downstream analyses, can be exported to `.csv` files for further visualization and statistical analysis.

- **From Window 2:** currently displayed I-T profile, current CV, all I-T profiles, all CVs
- **From Window 3:** peak amplitudes, area under the curve (AUC), I-T profiles, reuptake curves, fitting parameters

8.3 Exporting/Importing Sessions

Detected peaks, together with the pre-processing steps applied to them, can be exported from Window 2 and are included together with the data export from Window 3, allowing analyses to be recorded, shared, or reapplied to future datasets for consistency.

The export option in Window 2 also allows users to save an unfinished analysis, e.g. when working through a large dataset across multiple sessions.

Note: Exporting a session does not save experiment settings or replicate data (Sections 3 and 4.2), only pre-processing steps and detected peaks are included. Information on experiment settings and replicate data can instead be exported separately via the experiment log (Section 8.4).

Previous sessions can be re-imported after reloading the corresponding data, by clicking **Import Session** in the Export panel.

8.4 Exporting Experiment Log

An experiment log records the sequence of actions taken during a session, supporting reproducibility and record-keeping alongside the exported data. The log includes:

- **Experiment configuration**
- **Processing pipeline**
- **Data summary**
- **Replicate data**

The experiment log is exported together with all analysis metrics by clicking **Export Metrics as CSV** in Window 3.

9. Troubleshooting & FAQ

9.1 Common Loading/Import Issues

Note: NeuroStemVolt is designed for time-series data, so each replicate — with files corresponding to individual timepoints — needs to be loaded as a single folder (Section 4.2).

- Confirm the file format matches the supported format (Section 4.1).
- Verify that acquisition frequency and waveform settings match the original recording configuration (Section 3.1, 3.3).
- Check that file paths do not contain unsupported characters.
- Replicates with differing numbers of timepoint files (i.e., different numbers of files across replicate folders) are not supported for parallel/batch analysis. In this case, we recommend analyzing replicates individually and combining results afterward using the exported data.

9.2 Peak Detection Troubleshooting

- If peaks are missed or misidentified, revisit pre-processing settings (Section 5) before adjusting detection thresholds.
- Use manual review (Section 6.4) to correct individual events rather than re-running detection on the full dataset.
- Multi-Peak mode: if no peaks are validated, consider adjusting the detection parameters (Section 6.3).

9.3 Performance Considerations for Large/Batch Datasets

Large batch or replicate sets may take longer to process, particularly when multiple filtering steps are enabled. We recommend validating filter and peak detection parameters on a smaller batch first, before moving to large-scale analysis.

9.4 Platform-Specific Notes

- **macOS:** if the application will not open, verify Gatekeeper settings (Section 2.2.1).
- **Windows:** if SmartScreen blocks execution, use “Run anyway” (Section 2.2.3).

10. Appendix

A. Glossary of Terms

- **AUC (Area Under the Curve)** – the integrated area under a current–time profile or peak, exported as a metric for quantifying total signal/release over a given period (Section 8.2).
- **Background Subtraction** – a pre-processing step that removes non-faradaic (background) charging current by subtracting the mean signal from a defined baseline segment (Section 5.2).
- **Baseline Correction** – a pre-processing step that subtracts the mean current recorded prior to stimulation onset, so the profile begins at 0.0 nA (Section 5.6).
- **Butterworth Filter** – a low-pass filter used to attenuate noise while preserving waveform shape, controlled by filter order and cutoff frequency parameters (Section 5.4).
- **Color Plot** – a two-dimensional heat map representation of a single FSCV recording, with time on the x-axis, applied voltage on the y-axis, and current represented by color scale (Section 4.3).
- **CV Reference Line** – a manually set voltage line added to the CV and color plot views for visual comparison against a theoretical/expected oxidation peak voltage (Section 4.5).
- **Cyclic Voltammogram (CV)** – a plot of current against applied voltage, reproducing the characteristic voltammetric signature used to confirm analyte identity (Section 4.5).
- **Decay Constant (Tau)** – a parameter describing the rate at which a signal returns to baseline following a peak, extracted from exponential decay fitting and used to characterize reuptake kinetics (Sections 6, 7.2).
- **Fallback Mode** – a detection behavior in Single Peak mode in which the maximum signal value within a file is identified as the peak when no peak meets standard validation criteria (Section 6.2).
- **Files Before Treatment** – a treatment parameter defining the number of files/recordings collected prior to treatment addition, used to distinguish pre- and post-treatment files (Section 3.5).
- **FSCV (Fast-Scan Cyclic Voltammetry)** – an electrochemical technique for detecting rapid changes in neurotransmitter concentration.
- **HA Waveform** – the histamine-specific voltammetric waveform available in the waveform drop-down menu (Section 3.1; DOI: 10.1039/c5an00313j).
- **I-T Profile** – the current response over time, extracted from the color plot at the oxidation peak position (Section 3.4, 4.4).
- **Index** – the identifier used in the Edit Peaks pop-up window to label each detected or added peak by its position on the time axis (Section 6.4).
- **Jackson Waveform** – the serotonin (5-HT)-specific voltammetric waveform available in the waveform drop-down menu (Section 3.1; DOI: 10.1021/ac00102a015).
- **Min Height (nA)** – a Single Peak detection parameter setting the minimum absolute peak amplitude required for validation (Section 6.2).

- **Multi-Peak Mode** – an experiment mode in which multiple discrete events within a recording are identified and individually annotated (Section 6.3).
- **Non-Specific Waveform** – a waveform option that does not convert data points into voltages, allowing broader use of the software; disables CV visualization (Section 3.1).
- **Normalization** – scaling of detected peak amplitudes and profiles relative to the first file in a replicate's time series (Section 6.5).
- **Oxidation Peak Position** – the expected combination of voltage (and tolerance window) at which an analyte oxidizes, used to extract the I–T profile from the color plot (Section 3.4).
- **Prominence (%)** – a Single Peak detection parameter setting the required peak height relative to baseline (Section 6.2).
- **Replicate** – a folder of time-series data representing one experimental sample, containing one file per timepoint (Section 4.2).
- **Reuptake Curve** – the fitted decay-phase curve used to characterize the rate of neurotransmitter clearance following a peak (Section 7).
- **Rolling Mean Smoothing** – a pre-processing step that smooths a profile using a moving average over a sliding window of N points (Section 5.3).
- **Savitzky–Golay Filter** – a smoothing filter that fits a local polynomial across segments of the data, controlled by window size and polynomial order (Section 5.5).
- **Session (Export/Import)** – a saved record of pre-processing steps and detected peaks for a given analysis, which can be exported and later re-imported after reloading the corresponding data (Section 8.3).
- **Single Peak Mode** – an experiment mode in which one primary release event per recording is identified (Section 6.2).
- **Stimulation Reference** – a display overlay on the I–T profile marking stimulus start and duration, based on stimulation parameters and manually adjustable (Section 4.4).
- **Timepoint** – an individual recording (one file, one color plot) within a replicate's time series (Section 4.2).
- **Treatment Parameter** – the user-defined label describing the treatment or drug condition applied during an experiment (Section 3.5).

B. File Format Specifications

B.1 Input File Format

- **File type:** tab-separated `.txt` files (Section 4.1).
- **Structure:** columns = time points; rows = voltage sweep.
- **Folder structure:** one folder per replicate, containing one `.txt` file per timepoint (Section 4.2).

B.2 Exported Plot Files

Plots are exported as both `.png` and `.svg` files (Section 8.1).

From Window 2 (named by file ID, timepoint, and plot type):

- `[fileID]_[timepoint]_IT.png` — the current–time profile for the displayed file, titled with the detected peak's position (sample index). The peak amplitude is marked and annotated with its exact time (s) and current (nA) value.
- `[fileID]_[timepoint]_CP.png` — the full color plot for the displayed file, titled with the current color-scale range (min/max, in nA; Section 4.3).
- `[fileID]_[timepoint]_CV.png` — the CV extracted at the detected peak for the displayed file, titled with the source file name and annotated with the number of I–T peaks detected and the peak's voltage (V) and current (nA) values.

From Window 3 (Section 7):

- `mean_Its.png` (all modes, Section 7.1) — the mean I–T profile for every loaded file, overlaid on a single plot. Each file's profile is colored by treatment status (e.g., blue for files before treatment, green for files after treatment), with the first file after treatment highlighted separately for reference. The x-axis shows time points within the profile; the y-axis shows mean current (nA).
- `plot_mean_amplitudes_over_time.png` (all modes) — the mean normalized amplitude across replicates, plotted across timepoints, with a shaded SD band and a **Treatment Start** reference line marking treatment addition (Section 3.5).
- `plot_all_amplitudes_over_time.png` (all modes) — the normalized amplitude for each individual replicate, plotted across timepoints, with a **Treatment Start** reference line.
- `plot_tau_over_time.png` (Single Peak) — the exponential decay constant (τ) for each timepoint, plotted with error bars, with a **Treatment Start** reference line.
- `plot_exponential_fit_[timepoint]_[treatment].png` (Single Peak) — for the selected timepoint, plots the post-peak I–T decay data points (one per replicate) alongside the fitted exponential decay curve and its 95% confidence interval band, with a vertical reference line marking the fitted half-life ($t_{1/2}$). Timepoints are selected for export via a **Select Timepoints to Export** pop-up, which lists each by its timepoint label and source file, with **Select All** / **Select None** and **OK** / **Cancel** options.

- `spontaneous_peak_frequency.png` (Multi-Peak) — the mean peak frequency (peaks/min) across replicates, plotted across timepoints, with a shaded SD band, individual replicate traces, and a **Treatment Start** reference line.
- `peak_amplitudes_[window]s_windows.png` (Multi-Peak) — the mean peak amplitude averaged within user-defined time windows, plotted across the experiment, titled with the total number of peaks and windows included. Displays individual replicate traces alongside the mean, the number of contributing replicates (n=) labeled per window, and a **Treatment Start** reference line. The window size (in seconds) is set via an **Averaging Window** pop-up (default: 60 seconds) and is reflected in the exported file name.

B.3 Exported Data Tables

- `All_amplitudes_all_replicates.csv` — peak amplitude and peak position for every replicate, tracked across the experiment's timepoints. Uses a two-row header: the top row groups columns by metric (**Amplitude, Position (s)**), the second row labels each column by replicate (Rep1, Rep2, ...). Each subsequent row corresponds to one timepoint, indexed by the **Time** column (minutes).
- `All_AUC_all_replicates.csv` — area under the curve for every replicate, tracked across timepoints. Simple structure: a **Time** column (minutes) followed by one column per replicate (Rep1, Rep2, ...), one row per timepoint.
- `All_ITs_all_replicates.csv` — the full current–time (I–T) profile for every file in every replicate. Uses a two-row header: the top row groups columns by **Replicate**, the second row labels each column by its source **File** (e.g., 001.txt, 002.txt, ...). Rows correspond to sample points within the profile, indexed by **Time (s)**; values are current in nA.
- `All_ITs_all_replicates_paired_by_file.csv` — the same I–T profile data as above, reorganized so that columns are grouped by **File**/timepoint first and **Replicate** second. This layout pairs each replicate's profile at a given timepoint side by side, making it easier to compare replicates directly at the same point in the time course.
- `All_CVs_all_replicates.csv` — the cyclic voltammogram (current vs. voltage) extracted at the detected peak for every file in every replicate. Structured identically to `All_ITs_all_replicates.csv` (Section B.3): a two-row header grouping columns by **Replicate**, then by source **File**, with rows indexed by **Voltage (V)** and values representing current (nA) at each voltage step of the CV sweep.
- `All_reuptakes.csv` — the normalized decay (reuptake) profile following each detected peak, for every file in every replicate. Structured identically to `All_ITs_all_replicates.csv` (grouped by **Replicate**, then **File**), with rows indexed by **Time (s)** and values representing the normalized signal (starting at 1.0 at the peak).
- `All_reuptakes_paired_by_file.csv` — the same reuptake data as above, reorganized so that columns are grouped by **File**/timepoint first and **Replicate** second, mirroring the paired-by-file layout used for I–T profiles above.
- `exp_fit_joint.csv` — exponential decay fitting results, one row per timepoint (**Time_min**). Columns include the fit method; fitted amplitude and offset parameters (**A_fit**, **C_fit**) with standard errors and 95% confidence intervals; the decay rate constant (**k_fit_persec**) and its standard error; the derived decay constant (**tau_fit_s**).

and half-life (**t_half_s**), each with standard errors and confidence intervals; a fit status flag; and **n_used**, the number of data points used in the fit.

Note: Across all paired tables, replicates and timepoints are matched by column label (Rep1, Rep2, ... and file name, e.g. 001.txt), consistent file naming across replicates (Section 4.1) ensures these tables align correctly.

B.4 Exported Settings and Session Files

- **Session export (Section 8.3)** — creates a NeuroStemVolt-specific file, `peak_session.nsv`, containing pre-processing steps and detected peaks only (not experiment settings or replicate data).
- **Experiment log (`experiment_log_YYYYMMDD_HHMMSS.txt`, Section 8.4)** — a plain-text log containing:
 - Header with generation timestamp and output folder
 - **Experiment Configuration** — file type (Single/Multi-Peak), waveform, treatment, acquisition frequency, file length, oxidation peak position, time between files, and files before treatment.
 - **Stimulation Parameters** — start, duration, frequency, amplitude, and number of pulses.
 - **Calibration Settings**
 - **Processing Pipeline** — an ordered list of applied pre-processing steps and their parameters.
 - **Summary** — total files processed and source folders.
 - **Replicate Data** — per replicate, its metadata (treatment, waveform, number of timepoints, file length, acquisition frequency, time between files, files before treatment) followed by a per-file listing of file path, peak amplitude, peak position, baseline, and a flag indicating whether the peak was manually edited (Section 6.4).

C. Contact & Support Information

For technical support, questions, or to report issues, contact the Hashemi lab.